1. Operations Strategy

1.1. Operational Decisions

- Expanding or reducing capacity
- Locating a business or relocating it
- Offshoring or reshoring
- Outsourcing
- Changing operations (production) methods
- Application of IT and AI

The influence of human, marketing and finance resource availability on operational decisions

- Strategic operations decisions require assessment of human, marketing, and financial resources.
- If resources are insufficient, decisions may need to be revised or abandoned.

Some examples of resource dependency:

Expansion of Operation	Relocation of Operation
HR: Workforce planning	HR: Available labor supply in the new location
Marketing: Long-term sales forecasts	Operations: Efficient distribution system
Finance: Funds for expansion	Finance: Funds for relocation and site development
Inadequate Resources? Then, expansion plan may be abandoned.	Inadequate Resources? Then, alternatives must be considered (different site or staying at the current location).

Role of IT & AI in Operations Management

- IT has significantly expanded for service providers and product manufacturers.
- It enhances efficiency, automation, and decision-making.
- Examples: CAD (Computer Aided Design) & CAM (Computer Aided Manufacturing).

AI Applications in Business Operations:

- Automation & Productivity:
  - AI automates repetitive tasks, increasing efficiency.
  - AI-powered customer service can answer calls 24/7 and resolve 90% of issues on the first call.
- Data Processing & Decision-Making:
  - AI reduces search time and processes large amounts of data.
  - It enhances decision-making and boosts productivity.
- Employee Efficiency:
  - AI frees up employees for higher-level, creative tasks.
  - Improves operational efficiency and customer satisfaction.
- Banking:
  - Faster processing of loan requests.
- Online Payments:
  - AI detects and prevents fraud.
- Legal Sector:
  - AI identifies relevant past court cases.
- Manufacturing:
  - AI schedules maintenance for optimal efficiency.
- Pharmaceuticals:
  - AI predicts the success of drug development.
- Cybersecurity:
  - AI monitors behavior, detects threats, and responds to cyber risks.

Computer-Aided Design (CAD):

The use of computer programs to create two- or three-dimensional (2D or 3D) graphical representations of physical objects.

Key Usages of CAD:

Industry-Specific Uses	Product & Industrial Design	Engineering Applications
Architectural designs	Designing physical products (e.g. furniture, machinery)	Used throughout the engineering process
Graphic design as a final product	Creating detailed 3D models	Helps in component assembly analysis
Computer animation & special effects (movies, advertising, etc.)	Optimizing shape & size for efficiency	Aids in designing efficient manufacturing methods

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Benefits and Limitations of CAD:

Benefits of CAD	Limitations of CAD
Interactive & automated design analysis	Software can be expensive to purchase and maintain due to complex structure.
Minimizes errors, reducing material waste and production defects.	Employees must be trained to use the software efficiently as they need extensive training.
Reduces need for costly physical prototypes, lowering product development cost	Demands powerful hardware, increasing costs.
Improves design efficiency, manufacturing feasibility, enhancing precision and ensuring better design accuracy.	N/A
Reduces time to market for new products.	N/A
Provides clear 3D models of the final product and its components.	N/A

Computer-Aided Manufacturing (CAM):

The use of computer software to control machine tools and related equipment in the manufacturing of components or complete products.

- CAM automates factory production using robotic tools (e.g., lathes, milling, welding machines).
- Computers control each step, ensuring precision and consistency beyond human-operated machinery.

Benefits & Limitations of CAM:

Benefits of CAM	Limitations of CAM
High precision reduces quality issues in manufacturing.	High in cost due to expensive hardware, software, and employee training (though costs are decreasing).
Increased labor productivity and efficiency	Hardware failure/breakdowns can be complex and time-consuming to fix.
High flexibility allows quick product changeovers	Programming errors can cause faults that must be detected before production continues. This requires quality assurance.
CAD-CAM integration enables mass customisation, producing both niche and mass-market products on the same line	N/A
Helps businesses compete in small and large market segments, increasing market competitiveness	N/A

1.2. Flexibility & Innovation

Operation Flexibility

The ability of a business to vary both the level of production and the range of products following changes in customer demand.

It enhances efficiency by allowing firms to:

- Adapt Output to Market Demand: Businesses can scale production up or down based on fluctuations in customer demand, avoiding overproduction or shortages.
- Adjust Delivery Schedules: Companies can modify delivery timelines to align with changing customer needs, improving service reliability.
- Ability to do Unique Customer Specifications: Businesses can cater to specific customer requirements, offering customized products or services to maintain customer satisfaction.

How to improve Operation Flexibility:

- Increasing Capacity: Expanding facilities and acquiring more equipment enable businesses to handle larger volumes of production when needed.
- Outsourcing Production: Partnering with external suppliers allows firms to scale production quickly without long-term investments in infrastructure.
- Holding High Inventory Levels: Stockpiling raw materials or finished goods ensures ability to respond to sudden demand spikes.
- Employing a Flexible Workforce: A workforce that can adapt to different tasks or work schedules helps businesses adjust to operational needs efficiently.
- Investing in Mass Customization Systems: Advanced production technologies allow firms to offer personalized products while maintaining efficiency and cost-effectiveness.

Process Innovation:

The use of new or much-improved production methods or service delivery methods. It refers to the development and implementation of new methods of producing goods or delivering services. These innovations enhance efficiency, reduce costs, and improve the overall competitiveness of a business.

Examples of Process Innovation:

- Robots in Manufacturing: Automation reduces labor costs, increases precision, and enhances productivity in industries such as automobile and electronics manufacturing.
- Faster Microchip Manufacturing Machines: Advanced technology speeds up chip production, enabling faster and more powerful computers and electronic devices.

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- **Computerized Inventory Tracking:** Barcodes and scanners minimize stock shortages by providing real-time inventory updates, reducing lost sales and improving customer satisfaction.
- **Internet-Based Parcel Tracking:** Logistics companies track shipments globally, allowing customers to monitor deliveries and ensuring efficient supply chain management.

Benefits & Limitations of Process Innovation

Benefits	Limitations
Fall in Production Costs: Innovative techniques improve efficiency, reducing expenses related to labor, materials, and waste.	High Initial Costs: Investing in new technology, equipment, and training can be expensive.
Improves Competitiveness: Faster, more efficient processes enable businesses to respond quickly to market demands.	Resistance to Change: Employees may be reluctant to adapt to new processes, requiring time and effort for training.
Licensing Opportunities: Unique production methods can be patented and licensed, creating additional revenue streams.	Risk of Technology Failure: New processes may have technical issues, leading to disruptions and potential financial losses.

1.3. Enterprise Resource Planning (ERP)

Enterprise Resource Planning (ERP) is a software system that integrates all business functions (such as manufacturing, marketing, finance, and human resources) into a single platform. This enhances coordination, efficiency, and data accuracy across the company.

Benefits of ERP in Operations Management:

- **Improved Coordination:** ERP links all departments, ensuring seamless communication and data sharing.
- **Automation of Processes:** Automates inventory management, order processing, and financial tracking, reducing human errors.
- **Better Decision-Making:** Real-time data allows managers to make informed strategic decisions.
- **Cost and Time Savings:** Reduces redundancies, improves efficiency, and minimizes manual data entry.
- **Supply Chain Integration:** Connects suppliers and customers into a single network for smooth operations.

Efficiency Improvements from ERP Software:

- **Demand-Driven Supply Management:** Reduces waste by ensuring that materials and products are supplied only as needed, supporting sustainability.
- **Optimized Capacity Utilization:** Enhances workforce flexibility and enables quicker responses to demand fluctuations.
- **Just-in-Time (JIT) Inventory Management:** Orders materials only when required, reducing storage costs and waste.
- **Cost Reductions Across Supply Chain:** Electronically tracking materials and products minimizes inefficiencies and lowers costs.
- **Accurate Costing and Pricing:** Proper allocation of component and labor costs ensures precise pricing of products.
- **Improved Delivery Times & Customer Service:** Streamlined operations lead to faster order fulfillment and higher customer satisfaction.
- **Better Departmental Coordination:** A single database improves communication between departments, reducing errors and redundancies.
- **Better Decision-Making:** Real-time data from all business functions allows for informed strategic planning, such as selecting cost-effective suppliers.

It is beneficial for businesses as it enhances efficiency, improves customer service, and supports business-to-business (B2B) communication.

Limitations of ERP:

- **High Initial Costs:** Implementing ERP requires significant investment in software, hardware, and infrastructure, although costs are decreasing with technological advancements.
- **Employee Training & Resistance to Change:** Training employees is essential, as a poorly implemented system can reduce efficiency. Additionally, departments may resist adopting a unified system, preferring their existing methods.
- **Slow Implementation & Risk of Becoming Outdated**: Full ERP implementation can take 1–3 years, during which technology may evolve, making the system outdated. Upgrading ERP may require additional investment, potentially diverting funds from other important projects.

1.4. Lean Production

Producing goods and services with the minimum of wasted resources while maintaining high quality.

Aims and Purposes of Lean Production:

The primary goal of lean production is to produce high-quality output with fewer resources by eliminating unnecessary activities that do not add value to the customer. It includes decrease in:

- **Excessive Transportation:** Moving materials unnecessarily increases costs and inefficiencies.
- **Excessive Inventory:** Holding too much stock ties up capital and storage space.
- **Unnecessary Employee Movement:** Workers should not waste time retrieving materials.
- **Waiting Time:** Delays in production slow down efficiency.
- **Over-Production:** Producing more than demand leads to waste.

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- **Over-Processing:** Making products more complex than necessary increases costs.
- **Defects & Errors:** Poor-quality products require rework or disposal
- **Underutilized Talent:** Not leveraging employee skills reduces potential productivity.

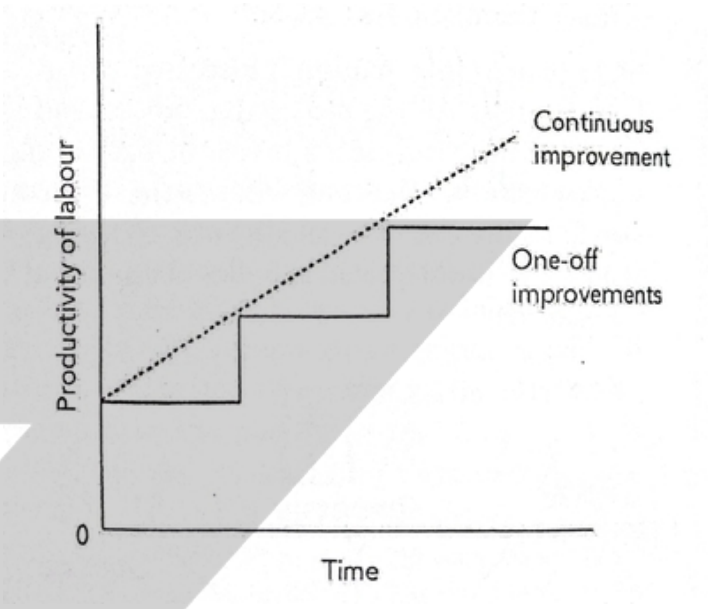
Key Lean Production Approaches:

- **Just-in-Time (JIT):** Producing goods only when needed to reduce inventory costs.
- **Total Quality Management (TQM):** Ensuring continuous quality improvement.
- **Employee Empowerment:** Encouraging workers to contribute to efficiency improvements.
- **Kaizen:** Japanese term meaning continuous improvement through small, slow and steady changes. It aims to encourage all employees.
- **Quality Circle:** It is a small group of employees who regularly meet to identify, analyze, and solve workplace problems to improve productivity and quality.
- **Simultaneous Engineering:** It is a method of product development where different stages, like design and production, are done at the same time to speed up the process and improve efficiency.
- **Cell Production:** It is a manufacturing method where workstations are arranged in a sequence, allowing teams to complete a product or component efficiently in a self-contained unit.
- **Waste Management:** Reducing the causes of waste in an organization. It is also collecting, recycling, and disposing of waste in an environmentally responsible way.

Kaizen's Core Aspects:

- **Employee Involvement:** Kaizen empowers all employees to identify and suggest improvements. It contains Theory Y managers. Those working directly with tasks often have practical insights into how to boost productivity and quality.
- **Small, Incremental Changes:** Instead of relying on large, one-off investments in new technology, Kaizen advocates for numerous small changes. Over time, these cumulative improvements can lead to significant gains in efficiency and quality.
- **Teamwork and Collaboration:** Improvements are typically discussed and developed within small groups or "Kaizen groups." These teams, drawn from the actual workforces, meet regularly to discuss problems, propose solutions, and even implement changes with the necessary managerial support.
- **System-wide Impact:** Changes made in one area often require adjustments elsewhere. For example, if a worker reduces the time needed to weld a car roof panel by five seconds, subsequent processes must also find efficiencies to prevent bottlenecks. This interconnected approach ensures that improvements are harmonized across the entire production process.

Visual Example of Kaizen:



Benefits and Limitations of Kaizen:

Benefits of Kaizen	Limitations of Kaizen
Continuous, small-scale improvements lead to a more efficient production process without the need for massive investments, boosting productivity.	Some challenges or improvements may require drastic, high-cost solutions (e.g., implementing an AI-based system) that cannot be addressed through gradual improvements.
Empowering workers by letting them contribute ideas boosts morale and leverages their expertise, leading to a more engaged workforce.	Authoritarian or traditional management styles may resist the empowerment and decentralization inherent in Kaizen, potentially hindering its effectiveness.
By streamlining processes and eliminating non-value-added activities, businesses can cut costs and reduce waste.	Initiating a Kaizen program often incurs training expenses and temporary productivity losses due to time spent in group discussions and meetings.
N/A	It is diminishing as most substantial gains usually occur early in the program. As the process continues, the incremental improvements might become less significant.

Quality Circle:

They are small groups of employees who voluntarily come together to identify, analyze, and solve quality-related problems within their workplace. Originating in Japan, this

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approach emphasizes teamwork, employee participation, and continuous improvement in production and service quality.

Key Aspects:

- **Employee Involvement:** Workers actively participate in problem-solving rather than just following management directives.
- **Regular Meetings:** Small teams meet periodically to discuss quality issues and suggest improvements.
- **Empowerment:** Some quality circles are given the authority to implement solutions without waiting for management approval.
- **Problem Identification & Solutions:** The main goal is to analyze quality-related issues and present effective solutions.

Benefits, Limitations, and Requirement for Quality Circle:

Benefits	Limitations	Requirement
Improves product and service quality through direct worker insights.	Weak management can hinder proper implementation.	Circle members must be committed to improving quality.
Boosts employee motivation and engagement by valuing their ideas.	Some employees may be reluctant to participate due to fear of criticism.	Members must be given training in holding meetings and problem-solving.
Encourages teamwork and communication across departments.	Lack of proper training can limit the effectiveness of discussions.	They need full support from management.
Reduces waste and inefficiencies in the production process.	N/A	Teams should be empowered to implement the recommendations.
Leads to continuous improvement (Kaizen) in workplace operations.	N/A	N/A

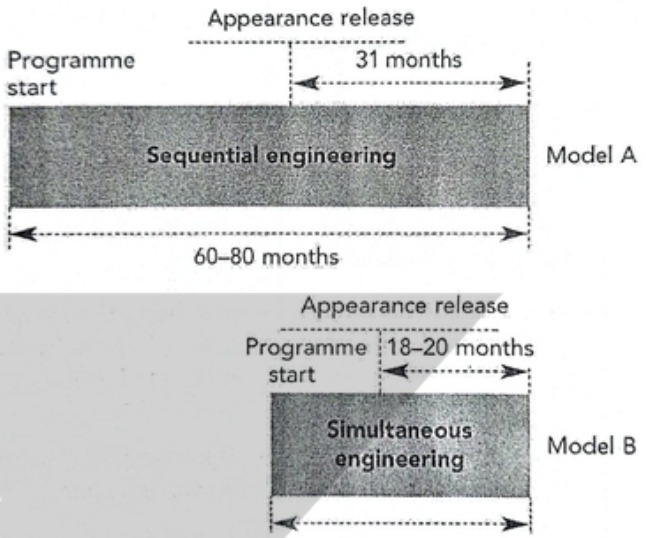
Simultaneous Engineering:

Simultaneous engineering, also known as concurrent engineering, is a product development approach where different stages of the process (such as design, market research, costing, and engineering) are conducted simultaneously rather than sequentially. This method significantly reduces the time needed to develop and launch new products.

Key Features:

- **Parallel Processing:** Multiple tasks are done at the same time, rather than waiting for one stage to finish before starting the next.
- **Cross-Functional Collaboration:** Teams from different departments (design, marketing, finance, and engineering) work together from the start.
- **Faster Time-to-Market:** New products reach consumers much earlier, giving businesses a competitive advantage.
- **Adaptability:** Reduces the risk of technological advancements making the product obsolete before launch.

Visual Example of Simultaneous Engineering:



Benefits and Limitations of Simultaneous Engineering:

Benefits	Limitations
Speeds up the time needed to launch a product.	Requires investment in collaboration tools and training.
Businesses can introduce new products ahead of competitors.	Teams must communicate effectively to avoid errors and poor coordination.
Early identification of design flaws avoids expensive late-stage modifications.	Employees may face higher workloads due to overlapping tasks.
Continuous feedback from different departments enhances product quality.	N/A

Cell Production:

Cell production is a type of flow production where the manufacturing process is divided into self-contained units, or "cells." Each cell consists of a team of multi-skilled workers responsible for producing a complete unit of work rather than just a small part. This approach enhances efficiency, quality, and worker motivation.

Key Features:

- **Self-Contained Units:** The production line is divided into independent "cells," each responsible for assembling a complete component (e.g., a washing machine motor).
- **Team Work:** Each cell consists of a small team of multi-skilled workers led by a team leader.

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- **Quality Responsibility:** Each cell ensures its own quality control, reducing defects and improving overall product quality (links with TQM).
- **Performance Targets:** Cells are measured against pre-set targets, including output, quality, and lead times.
- **Job Rotation & Multi-Skilling:** Workers switch tasks within the cell, increasing variety and engagement.

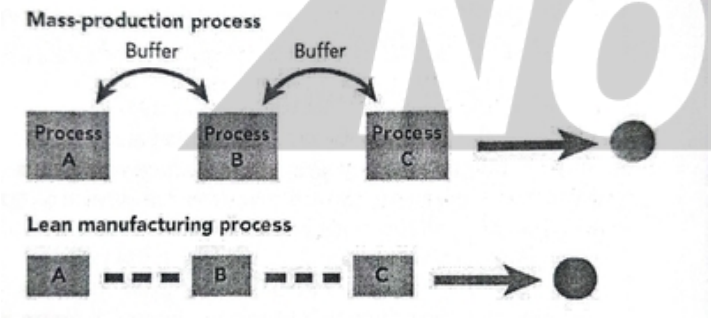
Benefits and Limitations of Cell Production:

Benefits	Limitations
Employees feel a sense of ownership over the final product, increasing motivation.	Employees need multi-skilling and training, which increases costs.
Encourages collaboration and communication within teams.	Shifting from traditional methods requires investment in layout changes.
Workers are accountable for quality control within their cell, leading to better quality control.	Best for moderate-to-complex manufacturing but impractical for mass production.
Reduces waste, lead times, and barriers in production, boosting efficiency.	If one cell underperforms, it slows down the entire process.
Easier to adapt to changes in demand and product design.	N/A

Just-in-Time (JIT) Inventory-Control's Key Aspects:

- JIT is a lean production strategy focused on reducing inventory and waste.
- It ensures that materials are delivered and products are produced only when needed, minimizing excess stock.
- The system operates with precise timing and coordination between suppliers, production, and distribution.

Visual Example of Just In Time Inventory Control:



Benefits and Limitations of JIT:

Benefits	Limitations
Minimizes the need for large storage areas and the costs associated with holding stock.	JIT relies on timely deliveries and can be vulnerable to supply chain disruptions, such as delays or shortages.
Resources are used more effectively, and there is less waste or unnecessary movement of goods.	Suppliers need to be highly reliable and able to deliver goods quickly, which can be a challenge.
The system is adaptable, allowing businesses to respond quickly to changes in demand or production needs.	Initial setup and training costs can be high as the system requires close coordination and advanced planning.
By minimizing the capital tied up in inventory, businesses can free up resources for other investments.	The lack of buffer stock means companies have less room for error in forecasting and planning.

Advantages & Disadvantages of Lean Production

Advantages	Disadvantages
Reduced Waste: Eliminates waste of time and resources, leading to improved sustainability.	Demand Forecasting Challenges: Difficulties in forecasting demand can cause issues, especially if there is an unexpected rise in demand.
Increased Efficiency: Boosts productivity, reduces average costs, and increases profits.	Customer Service Focus: Companies that rely on a wide range of products or guaranteed delivery might find lean production restrictive for meeting customer needs.
Improved Work Environment: Creates a less crowded and easier work area.	Job Losses: Increased efficiency can lead to redundancies, which may not align with the social objectives of certain businesses, particularly social enterprises.
Lower Inventory Risk: Reduces the risk of inventory damage due to lower stock levels.	High Costs of Technology and Training: The initial cost of implementing new technology and retraining employees can be prohibitive for some businesses.
Faster Product Launches: Speeds up time to market for new products.	Niche Manufacturers: Companies specializing in one-off or niche market products might find lean production unnecessary and may maintain competitiveness with traditional methods.
Improved Quality: Focus on continuous improvement enhances product quality.	-
Employee Motivation: Enriches jobs, enhances roles, and increases employee motivation.	-
Flexible Capacity Management: Allows better management of resources, especially in fluctuating demand conditions.	-

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Links Between Lean Production and Other Operations Issues

Inventory Control:

- Just-In-Time (JIT) inventory system helps in reducing waste and minimizing excess stock.

Quality:

- TQM (Total Quality Management) and zero-defects principles aim to reduce waste and enhance product quality.

Employees' Roles:

- The need for flexibility impacts employees' roles in three ways:
  - Flexible Employment Contracts: Non-core workers can be called in as needed.
  - Group Participation: Employees engage in groups (e.g., kaizen groups, quality circles) to suggest improvements.
  - Retraining: Employees are prepared to gain multi-skills to perform various jobs and work on different products.

Capacity Management

- Lean operations make better use of available capacity, reducing spare capacity by adapting designs for niche markets, which leads to higher demand and output.

Efficiency

- Lean production techniques improve efficiency by eliminating waste, lowering average production costs, and boosting competitiveness.

1.5. Operations Planning

Major operation decisions often involve major projects, such as relocation, expansion, and changes in operation methods.

Need for Planning Operations:

- Ensures efficient use of resources and minimizes waste.
- Reduces time taken to complete projects.
- Helps manage costs by avoiding idle resources and unnecessary expenses.

Examples of Business Projects:

- Setting up a new IT system.
- Relocating company operations.
- Installing new machinery.
- Expanding into international markets.
- Constructing new production facilities.

Resource Management in Operations Planning:

- Labor: Workers should only be employed when needed to avoid unnecessary wage costs.
- Materials: Just-in-time (JIT) delivery prevents excess inventory storage costs and reduces risks like theft.
- Machinery & Equipment: Specialist tools should be rented only for the required duration to manage costs effectively.
- Time Management: Efficient scheduling ensures minimal downtime and maximizes productivity.

Importance of Efficient Resource Use:

- Reduces costs and maximizes profitability.
- Prevents idle assets from increasing operational expenses.
- Ensures smooth workflow and timely project completion.
- Enhances cash flow management by avoiding unnecessary expenditures.

Critical Path Analysis (CPA)

A planning technique that identifies all tasks in a project, puts them in the correct sequence and allows for the identification of the critical path.

- Critical Path: The sequence of activities that must be completed on time for the whole project to be completed by the agreed date.
- Network Diagram: The diagram used in critical path analysis that shows the logical sequence of activities and the logical dependencies between them, so the critical path can be identified.

Key Aspects in CPA:

- Identify Project Objective: Define the goal, e.g., "Build a factory in six weeks."
- Sequence Tasks & Draw Network Diagram: Arrange activities logically.
- Estimate Durations: Assign time estimates to each task.
- Identify the Critical Path: Determine tasks that must be completed on time to meet the shortest project duration.
- Use the Network as a Control Tool: Monitor and adjust as needed.

Network Diagrams & Notation:

- Arrows represent activities (tasks).
- Nodes (circles) indicate the completion of activities.
- Activity durations determine the project's timeline.

Example: Building a House (3 Weeks Project)

- Critical Path Activities: A, C, E, F, H.
- Critical Path: If these activities are delayed, the entire project is delayed.
- Float Time: Some tasks (e.g., ordering the new machine) can have delays without affecting the project timeline.

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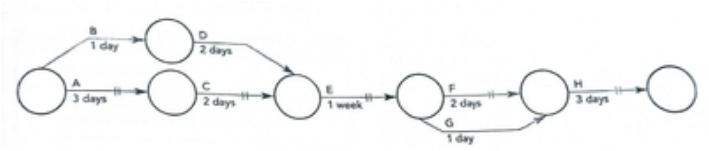


Benefits of CPA:

- Identifies the shortest completion time for a project.
- Helps allocate resources efficiently.
- Highlights critical activities requiring close monitoring.
- Provides flexibility through float time for non-critical activities.

How Critical Path Analysis is Determined:

- **List all Activities:** Break down the project into key tasks (e.g., digging foundations, building walls, tiling the roof).
- **Establish Task Dependencies:** Identify which tasks must be completed before others can begin (e.g., walls must be built before the roof is tiled).
- **Draw the Network Diagram:** Represent activities with arrows and use numbered nodes to indicate task completion.
- **Assign Durations:** Estimate the time required for each activity.
- **Identify the Longest Path:** The critical path is the sequence of tasks that determines the shortest possible project completion time.
- **Calculate Float Time:** Tasks not on the critical path may have float time, allowing flexibility in scheduling without delaying the overall project.

Order of Tasks:

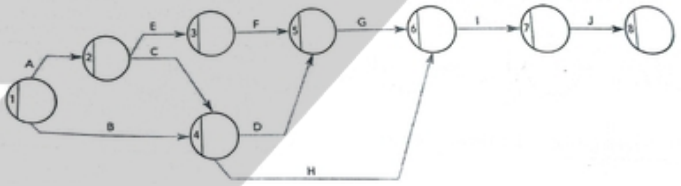
Activities	Preceding Activities
A	N/A
B	N/A
C	A
D	B & C
E	A
F	E
G	F & D
H	B & C
I	G & H
J	I

- **Earliest Start Time:** An activity cannot begin before this time.
- **Latest Finish Time:** An activity cannot finish later than this time without delaying a project.

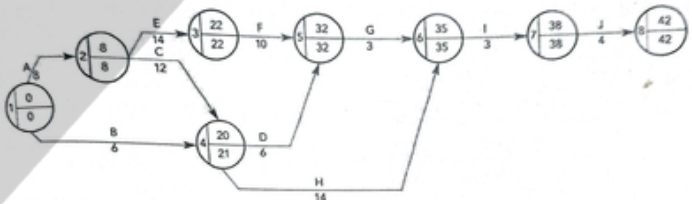
Duration Time of The Tasks:

Activities	Duration
A	8
B	6
C	12
D	6
E	14
F	10
G	3
H	14
I	3
J	4

Main Stages of The Project:



Inclusion of Activities & Durations of The Project:



Key Aspects of Critical Path Analysis (CPA)

Calculating the Earliest Start Time (EST):

- The EST is the earliest time an activity can begin, considering the completion of preceding tasks.
- Example:
  - Activity E cannot start before day 8 because Activity A must be completed first.
  - Activity D cannot start before day 20 as Activities A and C must be completed first.

Calculating the Latest Finish Time (LFT):

- The LFT is the latest an activity can finish without delaying the entire project.
- Example:
  - Activity J (final activity) must finish by day 42.
  - The LFT for Activity I is day 38 (42 minus the duration of J).
  - If multiple paths lead to a node, the lowest LFT is chosen to ensure minimal project delay.

Calculating Float Time for Non-Critical Activities:

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- Float time represents the spare time available for activities not on the critical path.

Types of Float:

- **Total Float:** The maximum delay an activity can have without delaying the entire project.
- **Free Float:** The delay allowed without affecting the next task in the sequence.

By carefully calculating EST, LFT, and float times, project managers can optimize schedules, allocate resources efficiently, and prevent bottlenecks/slowdown of the production.

For the example above:

The critical path ensures that the house is completed in 42 days. Delays in any critical activity will extend the project duration.

Key Features of Float Time and Critical Path:

Activity	Total Float	Free Float
B	15	14
C	1	0
D	6	6
H	1	1

Above is an example we will be using for the calculation.

Total Float:

- **Formula:** LFT - Duration - EST
- Example (Activity D):
  - LFT: 32
  - Duration: 6
  - EST: 20
- **Total Float Calculation:** 32 - 6 - 20 = 6 days
- **Meaning:** Activity D can be delayed by up to 6 days without affecting the project completion time.

Free Float:

- **Formula:** EST (next activity) - Duration - EST (this activity)
- Example (Activity B):
  - EST of next activity (H or D): 20
  - Duration of B: 6
  - EST at the start of B: 0
- **Free Float Calculation:** 20 - 6 - 0 = 14 days
- Meaning: Activity B can be delayed by 14 days without delaying the next activities.

Identifying the Critical Path:

- The critical path consists of activities with no spare time (where EST = LFT).
- Example (Node 3):
  - EST of F = 22, LFT of E = 22 → No delay possible.
  - Therefore, Activity F must start on time or the project will be delayed.
  - Critical Path: A → E → F → G → I → J
- Total Project Duration: 42 days

Importance of Float Times

- Total Float helps in resource allocation, allowing managers to adjust schedules without project delays.
- Free Float ensures that delays in one task don't affect subsequent tasks.
- Understanding float times prevents bottlenecks and allows for better contingency planning.

Dummy Activities:

A dummy activity is represented by a dotted line in a network diagram. It does not consume time or resources but shows logical dependencies between activities.

Key Features:

- It prevents the creation of illogical paths in network diagrams.
- It ensures accurate representation of dependencies between activities.
- It represents relationships, not actual tasks.
- It does not affect the project timeline or resources.

Example:

- Activities A and B start the project with no preceding activities.
- Activity C depends only on A.
- Activity D depends on both A and B.

Incorrect Network:

Below is an example, it incorrectly shows both C and D depending on A and B.

Correct Network:

- A dummy activity is used to indicate that:
- C starts after A is finished.
- D starts after both A and B are finished.
- The dotted line (dummy activity) links A to D to show this dependency.

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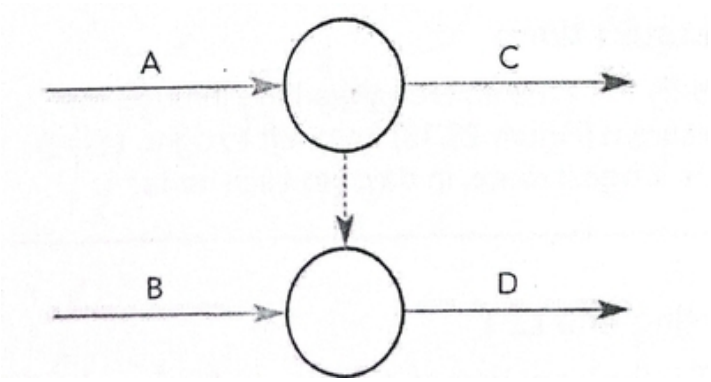

Benefits of Critical Path Analysis (CPA):

- Accurate Project Duration:
  - Helps calculate total project duration, allowing businesses to provide accurate delivery dates to customers.
  - Determines whether deadlines requested by customers can be met.
- Efficient Resource Ordering:
  - Early Start Times (EST) allow operations managers to order special equipment or materials at the right time.
  - Supports Just-In-Time (JIT) strategies and helps manage cash flow and working capital.
- Effective Control Tool:
  - Latest Finish Times (LFT) help monitor whether the project is on schedule by comparing actual completion times with network LFT.
- Critical Path Awareness:
  - Identifies activities with no float time where delays will impact the entire project.
  - Enables managers to speed up other activities to compensate for delays in critical tasks.
- Resource Optimization:
  - Resources from non-critical activities with float time can be allocated to critical ones to recover lost time.
  - Example: Delays in Activity E can be offset by reallocating resources from Activities D or H, which have spare time.
- Supports Computer Applications:
  - Logical and sequential structure of network diagrams is suitable for computer-based applications, enhancing efficiency and accuracy.
- Improves Planning:
  - Forces managers to carefully sequence activities and plan the project systematically.
- Simultaneous Engineering:
  - Highlights tasks that can be performed simultaneously during new product development.
  - Helps reduce the total time required for product development, benefiting fast-changing consumer markets.

Strengths, Drawbacks, and Requirements of CPA:

Strength	Drawbacks	Requirements
CPA aids in planning and controlling complex projects by visually identifying the sequence and duration of tasks.	CPA is less reliable for new projects with no prior data, as activity durations are often based on guesswork.	Successful implementation relies on skilled and experienced managers to use CPA effectively.
It provides a clear framework for identifying critical activities and float times.	Creating complex networks and calculating durations can take significant time and skilled labor.	Workers' commitment to the plan is critical. Involving employees in estimating activity durations can increase their buy-in and motivation.
It can help in optimizing resource allocation to improve efficiency and reduce delays.	N/A	Managers must identify cost-effective options to make up for delays in critical activities.
Useful for monitoring project progress and making adjustments when delays occur.	N/A	The cost and effort must be justified by the efficiency gains CPA provides.
N/A	N/A	A well-made CPA diagram cannot ensure project success without strong execution and management.

missing ^

- network diagrams as tools to plan operations
- the main elements of a network diagram: activities, dummy activities, nodes
- limitations of CPA after seperator (place indicated)
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